

Dr. Benjamin Rickles

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Professional Summary

Dedicated Cognitive Neuroscientist looking to apply skills to improve lives. Expertise in EEG data analysis, neurophysiological research, and statistical modeling. Proven ability to support clients in clinical settings and passionate about leveraging neurofeedback for cognitive and emotional wellbeing. Eager to apply analytical and interpersonal skills to optimize client outcomes and advance innovative neurotherapy practices.

Core Skills & Technologies

- EEG Data Acquisition & Analysis
 - Neurofeedback Protocols (Proficient in Alpha-Theta, SMR))
 - Python (Numpy, Scipy, Pandas, Scikit-learn, Jupyter notebook)
 - MATLAB & R (Statistical Analysis, Visualization, Data Processing, Signal Processing)
 - Machine Learning & Statistical Methods (Regression, Classification, Time Series Analysis)
 - Experimental Design & Human Subjects Research (IRB Protocols, Data Integrity)
 - Data Visualization (Effective Communication of Complex Physiological data)
 - Client Support & Communication (Empathy, Active Listening, Group Problem Solving)
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Research & Data Analysis Experience

Graduate Researcher | University of Maryland | 2017-2023

- Designed and implemented neurophysiological experiments, specializing in EEG and ERP studies to investigate cognitive and language processes.
- Developed and applied advanced signal processing techniques for EEG data preprocessing, including artifact removal, filtering, and feature extraction.
- Utilized Python, R, and MATLAB for statistical analysis, time-series analysis, and visualization of EEG data, identifying neural correlates of cognitive functions.
- Contributed to publications on neural oscillations and language comprehension

Research Intern | Adobe Inc. | 2020

- Developed algorithms for real-time neural data processing and integration of eye-tracking with EEG in virtual reality environments.
- Applied statistical models and machine learning techniques to analyze neurophysiological data, enhancing user experience and cognitive understanding.
- Visualized complex EEG data to provide intuitive insights and inform data-driven design decisions.

Additional Research Experience

- Conducted ERP/fMRI data analysis and experimental design across multiple institutions, focusing on neurocognitive modeling and improving reproducibility.

Professional Experience

Associate Clinical Account Specialist | Johnson & Johnson MedTech | 2024-Present

- Provided expert technical support during complex electrophysiology procedures, ensuring accurate data acquisition, troubleshooting technical issues, and optimizing data integrity.
- Collaborated closely with clinicians in high-pressure environments, facilitating seamless integration of technology into clinical workflows and improving patient outcomes.
- Excelled in client-facing roles, demonstrating strong communication skills, active listening, and a commitment to customer satisfaction.

Evaluation Coordinator | Advanced Education Research and Development Fund | 2023-2024

- Led scientific validation studies involving data collection and analysis for literacy interventions, focusing on cognitive and emotional factors influencing learning outcomes.
- Utilized statistical methods and data visualization tools to evaluate the effectiveness of educational programs and identify areas for improvement.

Chief Technology Officer & Innovator | Repurpose Farm Plastic LLC | 2021-2023

- Developed innovative recycling methods and secured research grants, demonstrating expertise in project management, problem-solving, and technical innovation.
- Managed stakeholder interviews, conducted research, and authored grant proposals showcasing strong communication and analytical skills.

Teaching & Leadership

- Instructor of Record for university-level courses on neuroscience, research methods, and developmental psychology.
- Led workshops in R-Studio, training students in data analysis tools relevant to neuroscience research.
- Mentored university students in research lab roles and in building and construction

Volunteer Experience

Instructor | Homeless Children's Education Fund | 2009-2014

- Met weekly with 1st-5th grade students after school to offer homework assistance

Awards & Recognitions

- Best Poster, 8th Annual John Hopkins Omega Psi Conference
- Hodos Award for Dissertation Research, University of Maryland
- Georgia State University Language & Literacy Initiative Graduate Fellowship
- University of Maryland Sustainability Grant for Research and Innovation